

CHAPTER 1

PREAMBLE

1.1 INTRODUCTION

Bikes & motorcycles are more popular among young people than four-wheelers, which is an established truth. At over 70% of bikers don't wear helmets, according to a poll. No apparent explanation why. We call any collision involving a motor vehicle that happens on road traffic accident. Tragically, the number of lives lost in incidents involving two wheelers is steadily rising. There is a significant loss of life in many of the incidents occurring around us. An estimated "7500" persons die each year in traffic accidents involving motorcycles, as per recent study. Serious accidents occur as a consequence of inattention, lack of training, and breaking traffic laws. Thus, our sense of civic duty to assist society in minimising accident occurrences motivated us to build our initiative. Inadequate driving skills, faulty two-wheelers, careless driving, "drinking and driving," and other similar offences are among many causes of accidents. But the major reason was person wasn't wearing helmet, resulting in brain damage and led to their instant death. Accordingly, a facility to lessen the impact of such incidents' aftermath must be in place.

3 main goals of a smart helmet are all practical for our daily lives. Most important thing is that we need to wear helmet before we can turn on bike. Secondly, it is not feasible to drive under influence while wearing a smart helmet. Bike will not start if rider is under influence of alcohol. Accident detection is the third application. These days, there are a lot of options for safety gear for bikers. System will consist of base unit installed onto bike & helmet module with stereo speakers & microphone. Helmet module will communicate wirelessly with bike, traffic signal, & speed breaker.

This system is operated by means of a speed sensor and FSR. Two circuits are communicating with one another via the use of radio waves. Due to small distance among circuits, a 315 MHz radio frequency module is utilised. A solar-powered smart helmet featuring several functions includes an engine control system, an integrated Bluetooth system, an accident alarm system, emergency alert switch, and the ability to charge a mobile device using solar energy.



Fig 1.1 : Road accident

1.2 OVERVIEW OF INTERNET OF THINGS

Users are able to accomplish greater system integration, evaluation, & automation with the help of IoT technologies. Both range and precision of these regions are enhanced by them. Sensors, networks, & robots are all part of the IoT, which makes use of both current and future technologies.

IoT takes use of evolving mindsets towards technology, dropping hardware costs, and new software advancements. Distribution of products, commodities, and services, as well as their social, financial, & political implications, undergo substantial transformations as a result of its innovative and cutting-edge features.

A network of interconnected physical objects that may exchange data with one another along with other networks with need for direct human intervention is known as IoT. "Internet of Things" refers, more precisely, to a network of interconnected computing devices which are able to exchange data with one another. Below, we will go over some of the potential concerns that Congress may have, including regulation, data security, and digital privacy.

1.2.1 IOT CHARACTERISTICS

Presence of sensors & ability to do sophisticated data analytics have earned IoT devices moniker "smart" gadgets. IoT devices employ sensors to gather data, which is then analysed in order to provide services to user, taking into account user-defined criteria. Take

smart refrigerator as an example. It employs sensors, like cameras, to keep track of what's in the fridge and may send out alerts when certain products are running short. Advanced Internet of Things devices have the ability to "learn" from their users' preferences and past actions. In order to increase user experiences or usefulness, an Internet of Things device might become "smarter" when its programme adapts to improve its prediction capabilities. Data exchange and user interaction are both facilitated by network connections. Through use of software applications, Internet of Things establishes links and connections among physical objects. Through the usage of various internet-connected devices, users of IoT devices may have knowledge of data or operate equipment remotely. If you have an intelligent doorbell & lock, for instance, you can see who's at door and even open it from your computer or mobile device—all without leaving your current location.

➤ **IOT– Key Highlights**

AI, connection, sensors, active involvement, & usage of tiny devices are most essential elements of IoT. Here is a quick rundown of these characteristics:

- AI – IoT basically renders almost everything "smart," which means it improves every part of human existence via data gathering, AI algorithms, & networks. A basic example of this would be programming your fridge & cupboards to send a signal to your favourite grocery store when supplies of milk and cereal are low.
- It is possible for networks to function with much reduced cost while maintaining substantially smaller size. IoT establishes these local networks among its device nodes.
- Without sensors, IoT becomes less distinctive. They are defining tools that make IoT more than just a network of devices; they make it active system that can be integrated with actual world.
- Passive Engagement - Passive engagement is the main mode of connection with linked technologies in modern times. Active interaction with information, products, or services is revolutionised by IoT.

➤ **IoT – Advantages**

- IoT has many practical and personal benefits given as:
- Better Customer involvement — As mentioned earlier, existing statistics have limitations while aren't always accurate, and involvement is still passive. The

Internet of Things revolutionises this, allowing for more effective and engaging audience participation.

- Optimisation of technology involves using same data along with technologies that enhance customer experience to make better use of devices and to make more powerful advancements in technology.
- IoT identifies areas that may be improved, leading to less waste. IoT data leads to better resource management than what we can get from current analytics, which only provide us surface-level understanding.
- Modern data collecting has its limits & is designed for passive usage, which may be improved with enhanced data collection. IoT removes it from those areas & puts it where people truly require it to go—to evaluate our reality. It provides a clear view of everything.

1.3 PROBLEM DEFINITION

Worldwide, road traffic accidents kill around 1.3 million people and damage an additional 20–50 million. The annual death toll from traffic accidents is still too high, at 1.24 million, as in 2013 Global Status Report on Road Safety. Thus, this smart helmet is being presented as a solution to this issue; it helps to decrease the number of accidents that occur daily and the fatality ratio.

Tragically, many individuals lose their lives in nations like India, wherein motorcycles are more common, because they fail to properly wear helmets. Lot of drivers still disregard repeated reminders from government officials to use safety gear, such as helmets & seat belts.

Traditional helmets are typically worn for sole purpose of warding off traffic control officers' challans, rather than for any real protection. Therefore, these headgear are not sufficient to protect driver. A helmet is an essential safety accessory for motorcyclists. Unfortunately, it can't tell you if rider is being very careful on road.



Fig 1.2 No of deaths due to road accidents

1.4 MOTIVATION

A smart helmet system in IoT (Internet of Things) integrates advanced technologies into traditional helmet designs, offering several key motivations and benefits:

- **Safety Enhancement:**

One of the primary motivations is to enhance safety for users. Smart helmets can incorporate features like impact sensors, GPS tracking, and emergency communication systems. These sensors can detect sudden impacts or accidents and immediately alert emergency services or designated contacts, ensuring timely assistance.

- **Real-time Monitoring:**

IoT-enabled helmets can monitor various parameters in real-time, like, moisture, & even biometric data like heart rate and fatigue levels. This data could be used to alert the wearer if conditions become unsafe or if they are exhibiting signs of exhaustion, helping prevent accidents.

- **Navigation and Communication:**

Integrated GPS and communication systems allow for easier navigation, especially in unfamiliar or hazardous environments. This is particularly useful for industries like mining, construction, and adventure sports where workers or athletes need to navigate complex terrains or remote locations.

- **Data Logging and Analysis:**

Smart helmets can gather dataset over time, that can be analyzed to identify patterns or trends related to safety incidents or environmental conditions. This data-driven approach enables organizations to implement proactive safety measures and optimize workflows.

- **Integration with IoT Ecosystems**

These helmets can be part of a larger IoT ecosystem within an organization or community. Could associate with other IoT devices such as drones, vehicles, or machinery, providing a comprehensive view of operations and enhancing overall safety and efficiency.

- **User Experience and Convenience**

Beyond safety, smart helmets can improve user experience and convenience. As instance, they could provide hands-free communication, voice commands for navigation, or even augmented reality displays for providing real-time instructions or data visualization.

- **Compliance and Regulations**

In some industries, smart helmets may be required to comply with safety regulations and standards. IoT functionalities can help assure it's maintained, reducing legal liabilities and enhancing workplace safety culture.

1.5 OBJECTIVE

Modern smart helmet system's primary function is identifying accident and notify control centre, ambulance, or victim's house in event that they need assistance. Without visible interaction from the control room, it can identify accidents and their severity. If this technology were integrated into all helmets, it would be simple to determine number of motorcycles involved in certain collision and the severity of it. In order for the control room's assistance to be in accordance with their standards. The current board design is more beneficial and practical since it includes both vehicles monitoring & accident alarm systems.

1.6 SCOPE OF PROJECT

In future we can use invisible camera to capture all the information regarding accidents which will also helps in future investigation.

- The camera in helmet will go inside when it's not in use because it can get damage when camera is outside.
- We could utilise solar panel instead of battery in helmet power.
- We could utilise set password or biometric for helmet if it gets stolen.

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORKS

Several comprehensive reviews about the subject of alcohol detection with smart helmets have been previously presented in the literature. Many such reviews focus on giving a global overview of the topic. Studies on alcohol detection systems include wearable helmets, mobile applications, and remote monitoring systems.

[1] The author in the year 2020 conducted an extensive review of the various communication and safety features integrated into smart helmets. The study highlights the evolution of smart helmets from basic protective gear to sophisticated devices capable of ensuring rider safety through advanced communication systems. The paper discusses the incorporation of Bluetooth technology for hands-free communication, GPS for location tracking, and GSM modules for emergency messaging. The authors emphasize the importance of these features in preventing accidents and providing timely assistance to injured riders. The study also explores the integration of sensors to monitor environmental conditions and the rider's physiological state, which can trigger alerts in case of abnormal readings.

[2] The author in the year (2021) delve into designing & implementing of smarter helmet system focused on real-time monitoring. The system utilizes various sensors, including accelerometers and gyroscopes, to detecting accidents & provide immediate alerts. It is equipped with a microcontroller that processes data via sensors and communicates with external devices via Bluetooth or Wi-Fi. The study emphasizes the importance of real-time monitoring in preventing severe injuries by ensuring that emergency services are notified promptly. Additionally, the authors discuss the challenges of integrating multiple sensors into a compact and lightweight helmet without compromising comfort or safety.

[3] The author in the yaer (2022) explore progress of smart helmet designed to detect alcohol consumption and prevent intoxicated individuals from riding motorcycles. The system incorporates alcohol sensor that detecting alcohol in the rider's breath. If alcohol

level exceeds a predefined threshold, the helmet communicates with the motorcycle's ignition system to prevent the engine from starting.

[4] The author in the year (2020) presents a smart helmet system capable of detecting accidents and automatically notifying emergency contacts. Helmet is equipped with sensors for detecting impact force during a collision, which triggers an alert to the rider's pre-configured emergency contacts through GSM and GPS modules. The system also includes a fall detection mechanism that activates when the helmet experiences a sudden drop, indicating a possible accident. The authors emphasize the importance of such systems in reducing the time between accident and the arrival of medical assistance, which is crucial for improving the survival rate of injured riders.

[5] The author in the year (2019) focus on the integration of multiple safety features into two-wheeler helmets using smart technology. Their research highlights the development of systems that can predict head injuries as per force of impact. If the system predicts a high risk of injury, it triggers a warning or deploys additional protective measures, such as inflatable padding. The study also discusses the potential for integrating augmented reality (AR) features into helmets, providing riders with real-time navigation information and hazard warnings without diverting their attention from the road.

[6] The author in the year (2021) provide current trends in smart helmet technology, focusing on future prospects. The study covers various aspects of smart helmet design, including sensor integration, power management, and user comfort. The authors highlight the need for advancements in battery technology to extend the operational life of smart helmets without increasing their weight. They also discuss the challenges of ensuring accurate sensor readings in different environmental conditions and the importance of user-friendly interfaces. The paper suggests that future developments in smart helmets will likely focus on improving user experience through better ergonomics and enhanced sensor accuracy.

[7] The author in the year (2020) explore the design of an IoT-based smart helmet system that ensures the rider's safety by detecting helmet usage and monitoring environmental conditions. The system includes sensors that detect whether the helmet is being worn and

communicate this information to the motorcycle's ignition system. If helmet is not worn, engine won't work. The helmet also monitors environmental conditions such as temperature and humidity, alerting the rider to potential hazards like extreme heat or rain. The study emphasizes part of IoT in enhancing the functionality of smart helmets, allowing for real-time data transmission and remote monitoring.

[8] The author in the year (2019) present a smart helmet system designed to prevent accidents through real-time monitoring and immediate accident detection. The helmet has sensors detecting the rider's head movements and predict potential collisions. If the system detects a high risk of an accident, it triggers an alert and activates preventive measures such as automatically adjusting the motorcycle's speed. It also has a fall detection feature that alerts emergency contacts in case of an accident. The authors discuss the potential of such systems to significantly reduce the number of motorcycle accidents and improve rider safety.

2.2 EXISTING SYSTEM

In the current framework The fundamental issue is engine vehicles created for improving human existence yet it impacts unfavorable on person as mishap. Street mishaps happen for the most part by engine vehicles and engine vehicle is just a human made gadget upon off chances which it causes gravely as street mishaps. Mishaps on streets make unsafe wounds the biker. Generally passing happens because of crash at mind of biker. This occurs due to keep away from the cap while driving. In the coincidental condition, essential treatment to the casualty is by and large postponed. This prompts expanding number of passing's in street mishaps. Street wellbeing is a critical concern around the world, with a huge number of individuals losing their lives or getting genuinely harmed in street mishaps every year. Among these, bicycle riders are especially defenseless because of the absence of actual security contrasted with vehicle drivers.

ADVANTAGES

- By preventing the motorcycle from starting if alcohol is detected, these systems significantly reduce the chances of accidents due to drunk driving.
- It can instantly detect alcohol levels in the rider's breath, providing real-time feedback

DISADVANTAGES

- GSN network is required.
- If helmet is stolen, the bike cannot be started.
- The sensor can detect every type of alcohol like deodorant, mouth freshener etc
- The project detects the alcohol only after certain level

2.3 PROPOSED SYSTEM

IOT-Based Smart helmet system for alcohol detection depicts the plan and execution of Brilliant Head protector utilizing IOT. There are two unique microcontrollers which are utilized in this undertaking. Every unit has utilized a different microcontroller, for bicycle unit we use Arduino Uno 2560 and for cap unit use Arduino Nano. The brilliant protective cap will have sharp IR sensor which will be utilized to distinguish rider has worn the head protector MD-3 sensor is operated for liquor location reason. Signal transmission between the protective cap unit and bicycle unit is utilizing a RF module nRF24L01 is used as RF module in our savvy cap nRF24L01 is chip radio for overall 2.4-25 GHz. It is in the transmitter side and other is utilized in recipient side. We have involved Arduino Nano in our Transmitter Circuit. At the point when Driver is tipsy. "BE Clearheaded AND WEAR Protective cap" is displayed in the OLED Display. The Engine Driver 1,298 is a double engine regulator and it can uphold 12V power supply as Source Voltage. In Collector circuit, we have utilized Arduino Mega 2560 as a microcontroller. DC Motor engine is utilized for 1.298.

Advantages

- **Enhanced Rider Safety:** The primary advantage is improved safety for the rider. By detecting alcohol levels, the helmet can prevent an intoxicated individual from starting the vehicle, reducing the risk of accidents due to impaired judgment and motor skills.
- **Accident Prevention:** The system can potentially reduce road accidents due to drunk driving. This not only protects the rider but also other road users.
- **Automatic Alerts:** Many smart helmet systems can be integrated with automatic alert mechanisms that notify emergency contacts or authorities if alcohol is detected. This can lead to timely interventions.

- Legal Compliance: drunk driving regulations, this system can help riders adhere to the law by preventing illegal riding under the influence.
- Insurance Benefits: Using a smart helmet with alcohol detection might lead to insurance benefits, such as lower premiums, due to the reduced risk of accidents.

CHAPTER 3

SYSTEM REQUIREMENT SPECIFICATION

They describe what system must do, how it should perform, and the constraints it must operate within. System requirements typically include:

3.1 FUNCTIONAL REQUIREMENTS

Describe what the system should do, its features, and functions.

3.1.1 SAFETY FEATURES

- **Impact Detection:** Sensors detect impacts or collisions, triggering alerts or emergency responses.
- **Fall Detection:** Algorithms detect sudden falls and can notify emergency contacts.
- **Helmet Integrity Monitoring:** Monitoring helmet condition to ensure it meets safety standards.

3.1.2 COMMUNICATION

- **Bluetooth/Wireless Connectivity:** Enables communication with smartphones or other devices for data exchange and alerts.
- **GPS Tracking:** Provides real-time location monitoring, useful in emergencies or for tracking routes.

3.1.3 USER INTERFACE

- **LED Indicators/Sound Alerts:** Provide feedback on helmet status, such as low battery or proximity warnings.
- **Voice Commands:** Hands-free operation for tasks like answering calls or adjusting settings.

3.1.4 DATA COLLECTION AND ANALYSIS

- **Environmental Monitoring:** Sensors for temperature, humidity, pollution levels, etc., providing contextual data.

- **Usage Analytics:** Collecting data on usage patterns and behavior for improving safety and user experience.

3.2 NON-FUNCTIONAL REQUIREMENTS

Describe how the system should perform, including performance, security, usability, and reliability.

3.2.1 ADVANCED COMMUNICATION FEATURES

- **Integration with Smart Assistants:** Voice-controlled interactions beyond basic commands.

3.2.2 AUGMENTED REALITY (AR) INTEGRATION

- **Heads-Up Display (HUD):** Overlaying information onto the helmet visor, such as navigation prompts or sensor data.
- **AR for Maintenance:** Providing real-time instructions or visual guides for maintenance tasks.

3.2.3 ADVANCED SENSOR INTEGRATION

- **Biometric Sensors:** Monitor heart rate, etc., for health-related insights.
- **Gesture Recognition:** Controlling functions through hand gestures for enhanced user interaction.

3.2.4 CUSTOMIZATION AND PERSONALIZATION

- **Customizable Alerts and Preferences:** Tailoring notifications and settings based on user preferences.
- **Integration with Wearables:** Syncing with smartwatches or fitness trackers for unified data monitoring.

3.2.5 CLOUD INTEGRATION AND DATA SECURITY

- **Cloud Storage:** Storing and accessing data remotely for backup or analysis.
- **Data Encryption:** Ensuring sensitive information is secure during transmission and storage.

3.3 SOFTWARE REQUIREMENTS

- Programming language: Embedded C
- Operating system: Windows 10
- IDE: Arduino IDE

3.3.1 EMBEDDED C

In order to resolve compatibility concerns between C extensions designed for various embedded systems, C Standards committee developed Embedded C, a collection of syntax extensions for compiler C. To handle exotic features like fixed-point arithmetic, many different memory banks, and fundamental I/O operations, embedded C programming has traditionally needed nonstandard extensions to the C language. To solve these problems and provide a uniform standard that all implementations should follow, the C Standards Committee expanded the C language in 2008. Features like named address spaces, basic I/O hardware addressing, and fixed-point arithmetic are not present in standard C.

3.3.2 WINDOWS 10 OS

Windows 10 will power an incredibly broad range of devices – everything from PCs, tablets, phones, Xbox One, Microsoft HoloLens and Surface Hub. It will also power the world around us, core to devices making up the Internet of Things, everything from elevators to ATMs to heart rate monitors to wearables. No matter which Windows 10 device you use, the experience will feel comfortable.

3.3.3 ARDUINO IDE (Compiler)

Text editor, a message area, a text terminal, toolbar containing buttons for common activities, & series of menus are all part of IDE, often known as Arduino Software. It establishes connection with Arduino hardware, allowing for programme uploading and communication.

1. Writing Sketches

Sketches are the programmes developed with the Arduino IDE. Such drawings are penned in text editor and then saved with .ino extension. When you save or export, the message

section will tell you what happened and any mistakes that occurred. Everything from full error messages to other information is shown on console by means of text output via IDE. You may launch serial monitor, create, open, & save drawings, & check & upload programmes using buttons upon toolbar.

- Verify Checks your code for errors compiling it.
- Upload Compiles your code and uploads it to the configured board. See uploading below for details. Note: If you are using an external programmer with your board, you can hold down the "shift" key on your computer when using this icon. The text will change to "Upload using Programmer"
- New Creates a new sketch.
- Open Presents a menu of all the sketches in your sketchbook. Clicking one will open it within the current window overwriting its content.
- Additional commands are found within the five menus:
- File, Edit, Sketch, Tools, Help. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

2. File

- New Creates a new instance of the editor, with the bare minimum structure of a sketch already in place.
- Open Allows to load a sketch file browsing through the computer drives and folders.
- Open Recent Provides a short list of the most recent sketches, ready to be opened.
- Sketchbook Shows the current sketches within the sketchbook folder structure; clicking on any name
- Page Setup It shows the Page Setup window for printing. Print Sends the current sketch to the printer according to the settings defined in Page Setup.
- Preferences Opens the Preferences window where some settings of the IDE may be customized, as the language of the IDE interface.
- Quit Closes all IDE windows. The same sketches open when quit was chosen will be automatically reopened the next time you start the IDE.

3. Edit

- ▮ Return to Previous Action Goes back a few steps you took if editing; you can keep using Redo after you go back.
- ▮ Copying and pasting makes copy of current selection and pastes it in clipboard.
- ▮ Paste for Forum Pastes your sketch's code onto clipboard in format that's appropriate for sharing on forum, including syntax colouring.
- ▮ Cut and Paste HTML Copies your sketch's code to clipboard in an HTML format that may be easily embedded onto websites.
- ▮ Rise or Fall Indentation moves text one space to right or removes a space from beginning of each chosen line by adding or subtracting space at beginning of the line.
- ▮ Find & Replace box opens, allowing you to begin searching for text inside current drawing based on several settings.

4. Sketch

- ▮ Using specified Port, the programme compiles & transfers binary file to specified board.
- ▮ Programmatically Upload This will replace board's bootloader.
- ▮ Use .hex file which is saved when you choose "Export Compiled Binary" to either send it to board or save it as an archive.
- ▮ Display the Sketch Bundle Activates the sketch folder currently open.
- ▮ Develop a library Inserts #include lines into your code at beginning of your sketch, adding a library to it. Check out the library down below for all the info.
- ▮ Selecting "Add File..." copies file's present location and adds it to drawing. Documents and other materials should go in data subdirectory of drawing, where file is stored.

5.Tools

- Auto Format This formats your code nicely: i.e. indents it so that opening and closing curly braces line up, and that the statements inside curly braces are indented more
- Archive Sketch Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.

- **Fix Encoding & Reload** Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
- **Serial Monitor** Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.
- **Board** Select the board that you're using. See below for [descriptions of the various boards](#).
- **Port** This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu.

6. Sketchbook

Arduino IDE is based on idea of a sketchbook, which is common location to save your programmes and designs. To access drawings saved in your sketchbook, go to File > Sketchbook or use Open button located on the toolbar. Arduino software will generate directory for your sketchbook moment you launch it for first time. Within Preferences dialogue, you have option to either see or modify exact spot of sketchbook. Workspaces, Various Files, & Assembly Gives you the option to work with many files at once, allowing you to easily organise your drawings. You may find standard

Standard Arduino code files (.ino,.pde) for sketch are joined into one file for tabs appear before sketch is compiled. No changes are made to other file formats.

7. Uploading

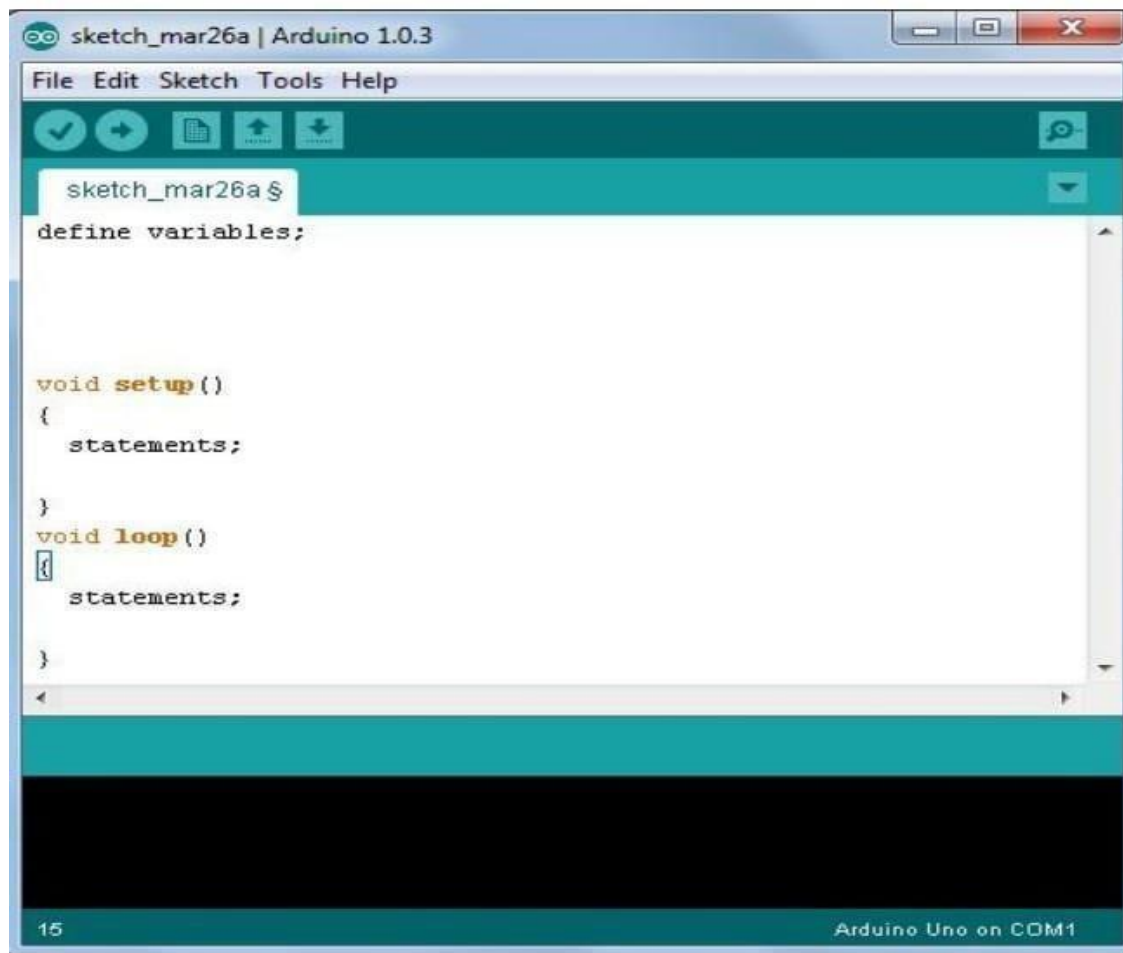
Before uploading your sketch, you need to select the correct items from the **Tools > Board** and **Tools > Port** menus. On Windows, it's probably **COM1** or **COM2** (for a serial board) or **COM4**, **COM5**, **COM7**, or higher (for a USB board) - to find out, you look for USB serial device in the port section of the Windows Device Manager. On Linux, it should be **/dev/ttyACMx**, **/dev/ttyUSBx** or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the **Upload** item from the **Sketch** menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

8. Serial Monitor

See the interfaces page for instructions on how to communicate with the board using Processing, Flash, MaxMSP, and other similar programmes.

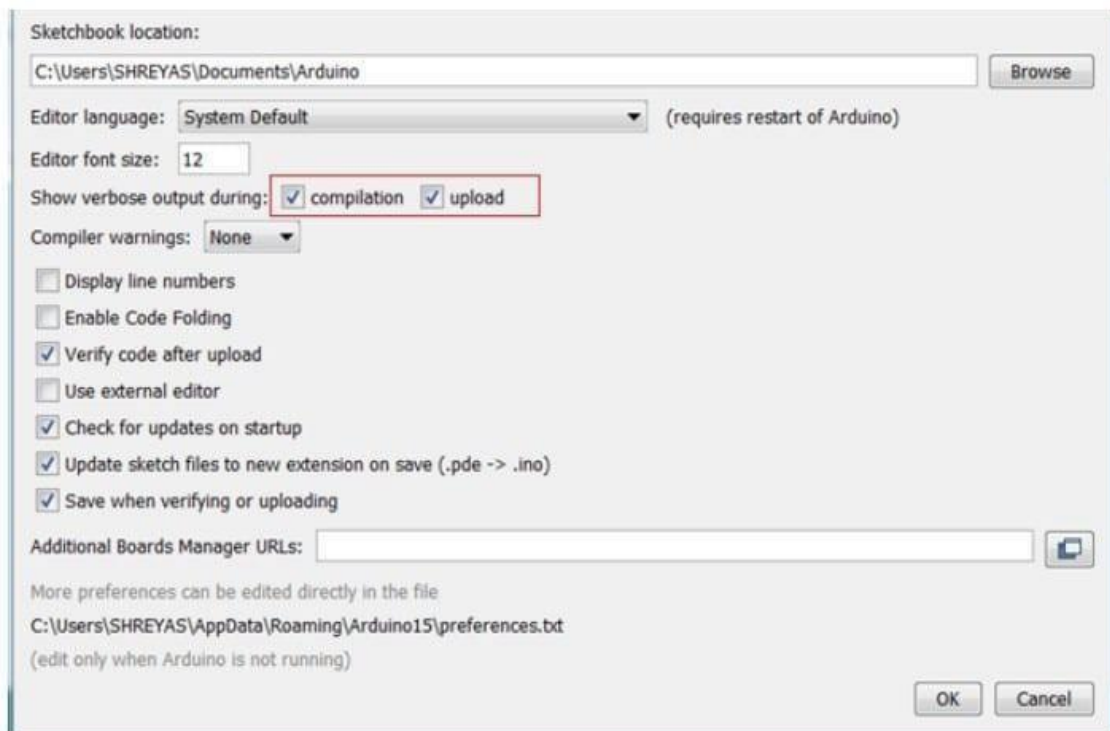
9. Preferences

You can adjust several settings in preferences dialogue, which is accessible from the Arduino menu on Macs or File on Windows and Linux.

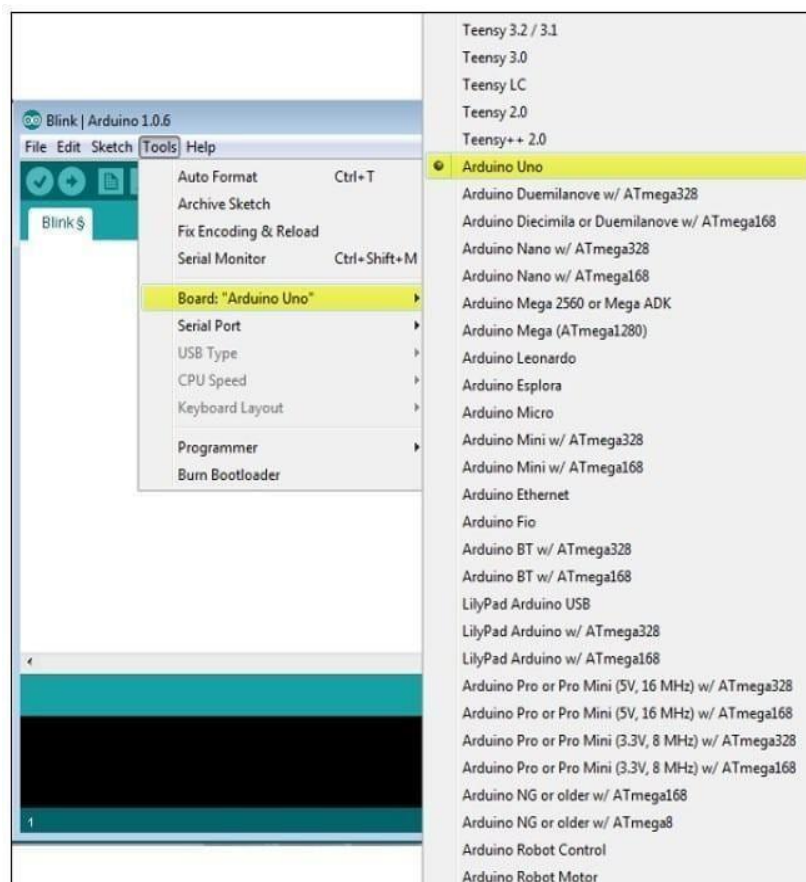


Arduino IDE Software Steps

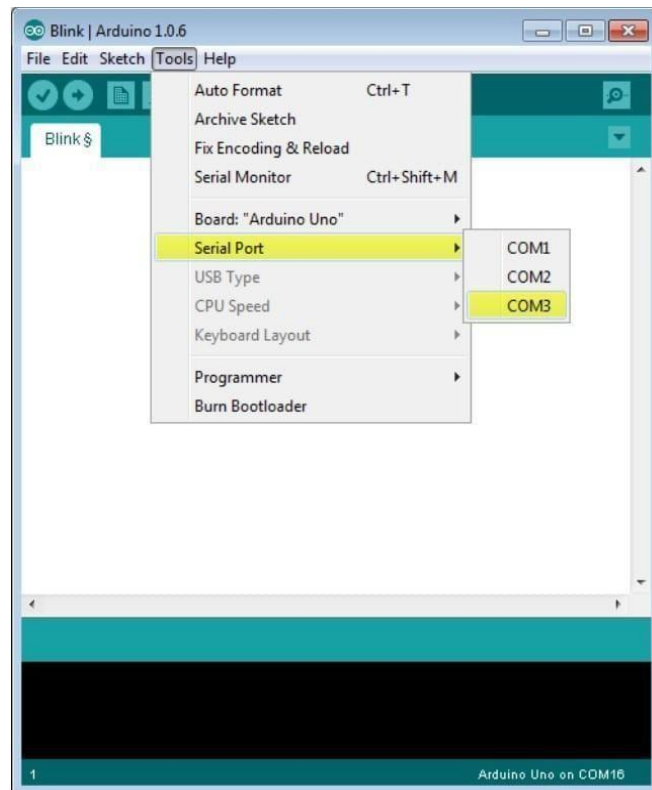
1. Start the Arduino editor & from File menu select Preferences to display the screen shown below. Find and select the option to 'Show verbose output during;' and tick the 'compilation' checkbox. With this option selected, the Editor will show a trace of each step of the compilation process.



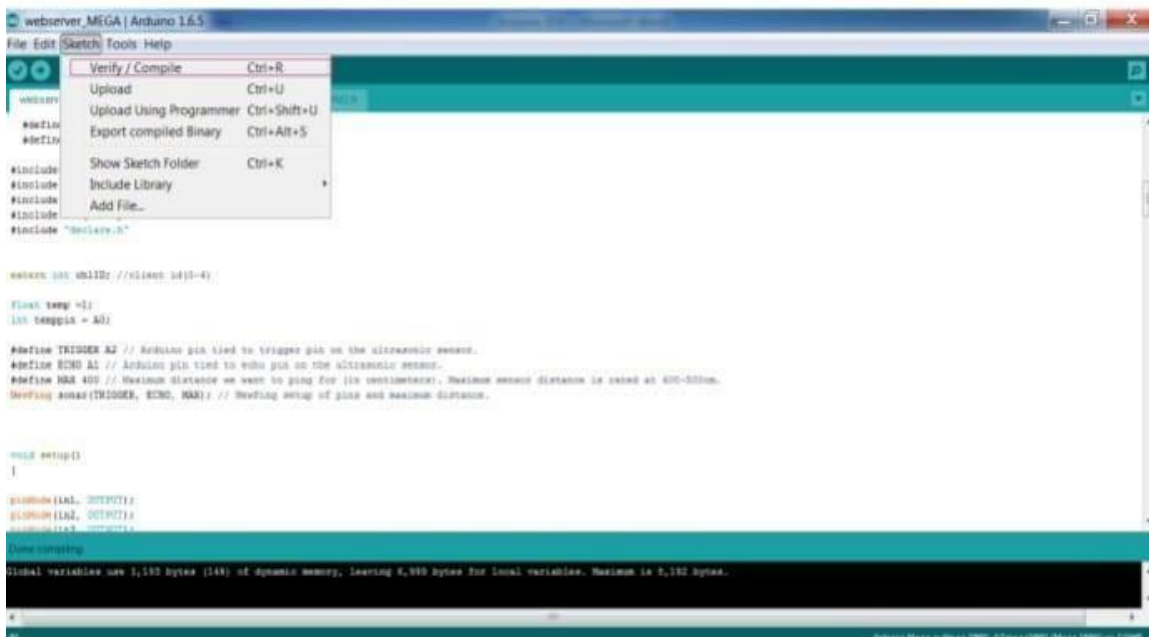
2. Open your source code file and select your Board from Tools Menu



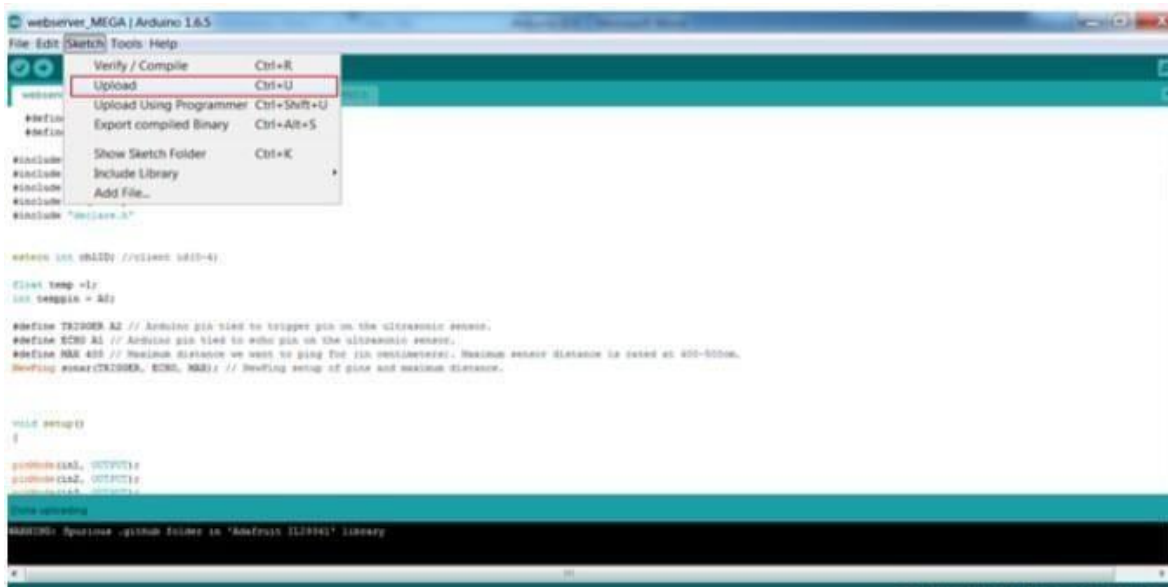
3. Select available Port from Tools Menu.



4. Click Verify/ Compile from Sketch Menu. If there are no errors in the code, it shows Done Compiling at bottom side.



5. Now click Upload from Sketch Menu. After uploading the code, it shows Done Uploading at bottom side. Now board is ready to use in hardware application.



6. You have successfully compiled and uploaded program code to microcontroller.

3.4 HARDWARE REQUIREMENTS

Describe the hardware components, configurations, and interfaces.

- Arduino Uno: The focal microcontroller unit for the undertaking.
- Liquor Sensor (MQ-3): To identify liquor in the rider's breath.
- Switch: For assuring head protector is worn accurately.
- GSM Module (SIM900): For sending SMS alarms if there must be presence of crises.
- GPS Module (NEO-6M): For ongoing area following.
- Ringer: To give discernible alarms.
- Protective cap: The actual cap wherein all parts will be incorporated.
- Power Supply : Batteries or other power sources for controlling framework.

3.4.1 ARDUINO UNO

The Arduino Uno is an open-source microcontroller board that depends on the Microprocessor ATmega328P (for Arduino UNO R3) or Computer chip ATmega4809 (for Arduino UNO WIFI R2) miniature regulator by Atmel and was the principal USB

fueled board created by Arduino. The Arduino Uno R3 with Link is a microcontroller board in view of the AT-mega (datasheet). It has 14 computerized input/output pins (of which 6 can be utilized as PWM yields), 6 simple sources of info, a 16 MHz earthenware resonator, a USB association, a power jack, an ICSP header, and a reset button. It contains everything expected to help the microcontroller, just interface it to a PC with a USB link or power it with an air conditioner to-Dc connector or battery to begin. "Uno" signifies one in Italian and was decided to check the arrival of Arduino Programming (IDE) 1.0. The Uno board and form 1.0 of Arduino Programming (IDE) were the reference variants of Arduino. The Uno board is the principal in a progression of USB Arduino sheets, and the reference model for the Arduino stage.

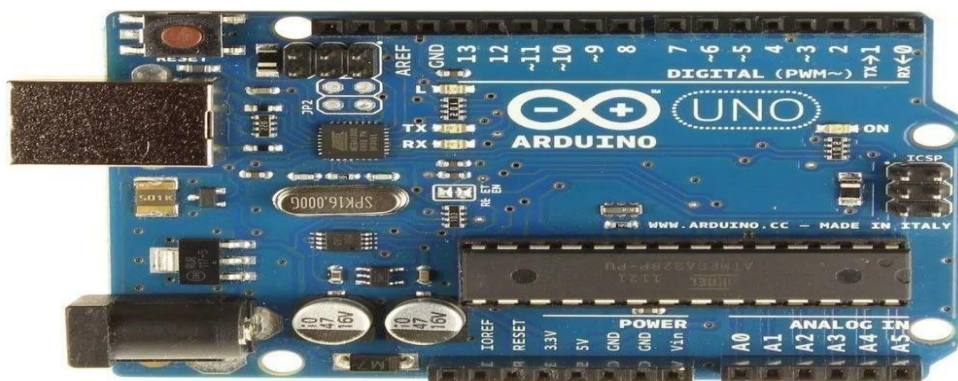


Fig 3.1 Arduino Uno

3.4.2 MQ3 SENSOR

The MQ-3 sensor is a sort of gas sensor that is generally used to recognize the presence of liquor fume in the air. It is explicitly intended to be delicate to liquor, making it helpful in applications where liquor discovery is vital, for example, breathalyzer gadgets or liquor Checking frameworks.

Working Guideline: The sensor works on the standard of a chemiresistor, and that implies that its electrical obstruction changes within the sight of the objective gas.

The MQ-3 sensor contains a tin dioxide (SnO_2) detecting component that cooperates with the liquor particles. At the point when liquor fume is available it responds with the detecting material, causing an adjustment of its obstruction. This adjustment of opposition can be estimated and related to the grouping of liquor in the air.

The MQ-3 sensor requires a warming component to successfully work. The warming component raises the sensor's temperature, working on its awareness and reaction time. Regularly, the sensor requires a consistent power supply and a warm-up time before precise readings can be gotten.

Determinations: The MQ3 liquor sensor works on 5V DC and consumes roughly 800mW. It can distinguish liquor focuses going from 25 to 500 ppm.



Fig 3.2: MQ3 Sensor

3.4.3 IR SENSOR

The functioning guideline of IR sensor is like the article discovery sensor. This sensor incorporates an IR Drove and an IR Photodiode, so by consolidating these two can be framed as a photocoupler in any case optocoupler. The material science regulations utilized in this sensor are boards radiation, Stephan Boltzmann and weins removal. IR Drove is one sort of transmitter that produces IR radiations. This Drove seems to be like a standard Drove and the radiation which is produced by this isn't noticeable to the natural eye. Infrared beneficiaries essentially identify the radiation utilizing an infrared transmitter. These infrared beneficiaries are accessible in photodiodes structure. IR Photodiodes are disparate as contrasted and common photodiodes in light of the fact that they distinguish just IR radiation. Various types of infrared beneficiaries predominantly exist contingent upon the voltage, frequency, bundle, and so on. Whenever it is utilized as the mix of an IR transmitter and recipient, then, at that point, the beneficiary's frequency should rise to the transmitter. Here, the transmitter is IR Driven while the collector is IR photodiode. The infrared photodiode is receptive to the infrared light that is created through an infrared Drove. The

opposition of photodiode and the adjustment of result voltage is with respect to the infrared light acquired. This is the IR sensor's central working guideline. When the infrared transmitter produces emanation, then it shows up at the article and a portion of the discharge will reflect back toward the infrared beneficiary. The sensor result can be concluded by the IR collector relying upon the force of the reaction.

IR Sensor Module Particulars and Elements 5VDC Working voltage I/O pins are 5V and 3.3V agreeable Reach: Up to 20cm Flexible Detecting range Worked in Encompassing Light Sensor 20mA stock current

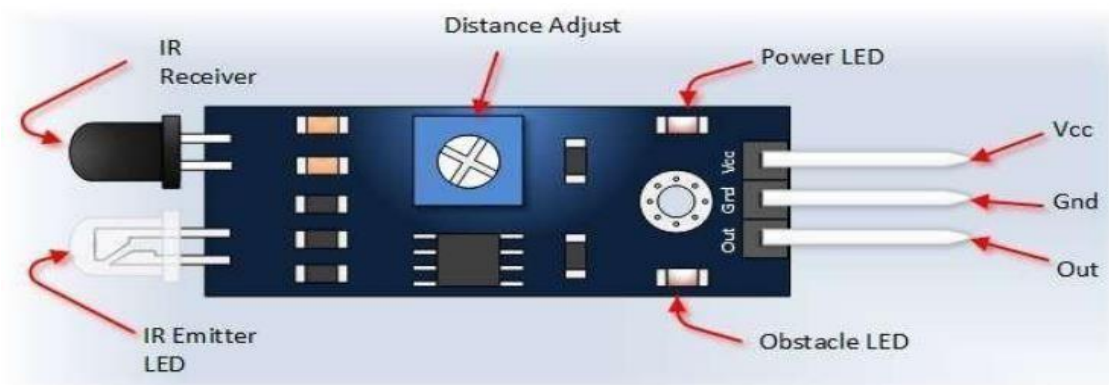


Fig 3.3: IR Sensor

3.4.4 16*2 LCD Show

A 16x2 LCD show alludes to a sort of fluid gem show (LCD) usually utilized in electronic gadgets. The "16x2" alludes to the showcase's aspects, demonstrating that it could portray show 16 characters evenly and 2 columns in an upward direction. These presentations are basically liked for multi portion light emanating diodes and seven sections. The primary advantage of utilizing this module are reasonable; essentially programmable, livelinesss ,and there are no limits for showing customs character, exceptional and even movements ,and so on Highlights:

The elements of this LCD essentially incorporate the accompanying.

- The working voltage of this LCD is 4.7V-5.3V.
- It Incorporates two lines where each column can create 16 characters.
- The use of current is 1mA with no backdrop illumination.

Each character can be worked with a 5*8 pixel box.

- The alphanumeric LCDs letter sets and numbers.
- Is show can chip away at two modes like 4-digit and 8-bit.
- These are obtainable in blue and green back drop illumination.

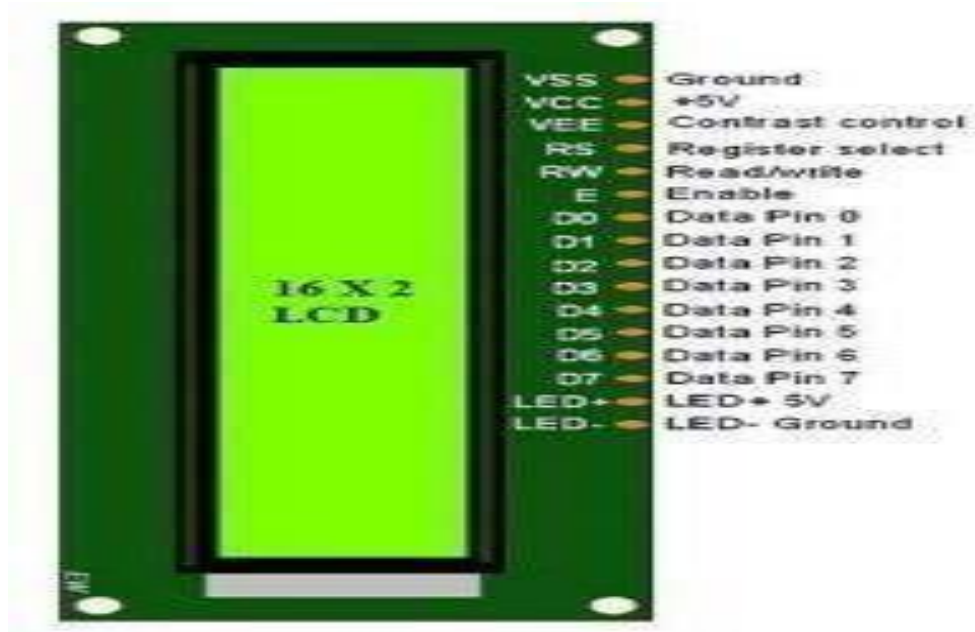


Fig 3.4 16*2 LCD Show

3.4.5 JUMPER WIRES

Jumper wires are basically wires that have connector pins at each end, permitting them to be utilized to interface two focuses to one another without welding Jumper wires are regularly utilized with breadboards and other prototyping devices to make it simple to change a circuit on a case by case basis

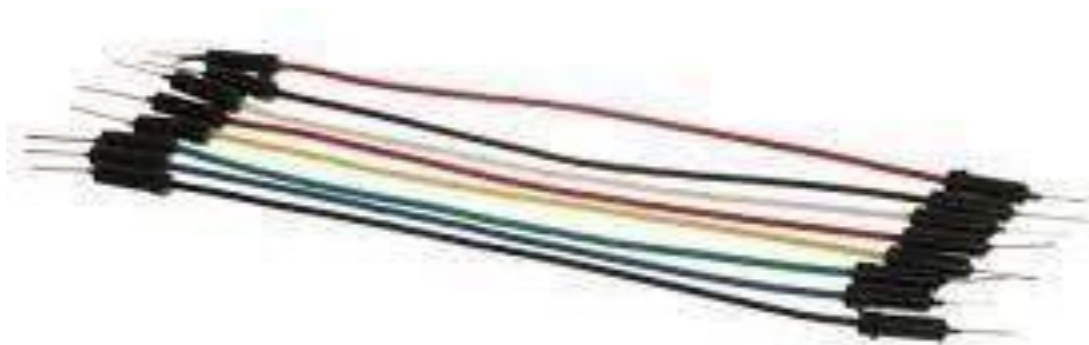


Fig 3.5 Jumper Wires

3.4.6 MICROCONTROLLER ATMEGA328

Arduino UNO is an open source prototyping stage in light of ATmega328 microcontroller. It comprises of 14 computerized input/output (I/O) pins, six simple data sources, a USB association for programming the on-board microcontroller, a power jack, an ICSP header and a reset button. It is worked with a 16MHz gem oscillator and contains everything expected to help the microcontroller.



Fig 3.6 Microcontroller Atmega328

3.4.7 ALCOHOL SENSOR

There is vast variety of liquor sensors available, each with its own set of advantages and disadvantages depending on factors like responsiveness, liquid type to be detected, physical characteristics, and more. An understanding of methane gas sensors which may detect gases, such as odorant salts, that might be produced by methane, is included in this understanding.



Fig 3.7 Alcohol Sensor

3.4.8 GPS AND GSM MODULE

The Worldwide Situating Framework (GPS) is a satellite-based route framework that is used to identify where the mishap will be occurred. It recognizes Longitude and Scope upsides of specific spot and sends it to GSM module. It works in every single atmospheric condition. It additionally assists with deciding different units like speed, distance, time, and so on.



Fig 3.8: GPS And GSM Module

3.4.9 TRANSFER

One way to electrically separate & connect two circuits is via a transfer, an electromagnetic device. These handy devices enable a single circuit to switch other even if they are separate. Connecting 2 electrical circuits, one operating at low voltage and the other at very higher voltage, is a common purpose for them.

3.4.10 DC ENGINES

A DC engine is a brushless, coordinated electric engine that can partition a full pivot into countless strides, for instance 200 stages. In this way the engine can be gone to an exact point. Computerized signal is utilized to drive the engine and each time when it gets an advanced heartbeat point. Computerized signal is utilized to drive the engine and each time when it gets a computerized beat it turns a particular number of degrees.



Fig 3.9: DC Engines

3.4.11 SWITCH/BUTTON

press button The button goes regarding model used to turn over the motor of a vehicle. Press Buttons are generally open material switches. Press buttons permit to make a specific association when the button is squeezed. It essentially makes the circuit associated when squeezed and breaks whenever its delivered.



Fig 3.10: Switch

3.4.12 BELL

A electromechanical ringer or bleeper is a device that flags sounds. Alarms, clocks, and client information confirmation are common applications for bells and beepers. One possible source of driving a piezoelectric component is a pulsing electrical circuit or some other kind of audible signal. Typically, a tick, bell, or signal will be utilised to show when button has been pressed.



Fig 3.11:Bell

CHAPTER 4

SYSTEM DESIGN AND DEVELOPMENT

4.1 INTRODUCTION

The smart helmet system for alcohol detection is design and development of the Arduinobased smart helmet project. The system integrates various components, such as an alcohol sensor, a switch, a buzzer, a GSM module, and a GPS module, to ensure the safety of motorcyclists. The primary aim is to prevent drunk driving and ensure timely assistance in case of an accident.

4.1.1 OVERALL SYSTEM ARCHITECTURE

The helmet section has touch sensor which gives the information about the status of whether the helmet is occupied or not, also we have sensors like gyroscope and vibration sensor for knowing position of accident. Such dataset fetched from the helmet section is transferred through RF technology and reception take place at the vehicle section. The data is also update to cloud so that we are able to either control or monitor the respective system using IOT.

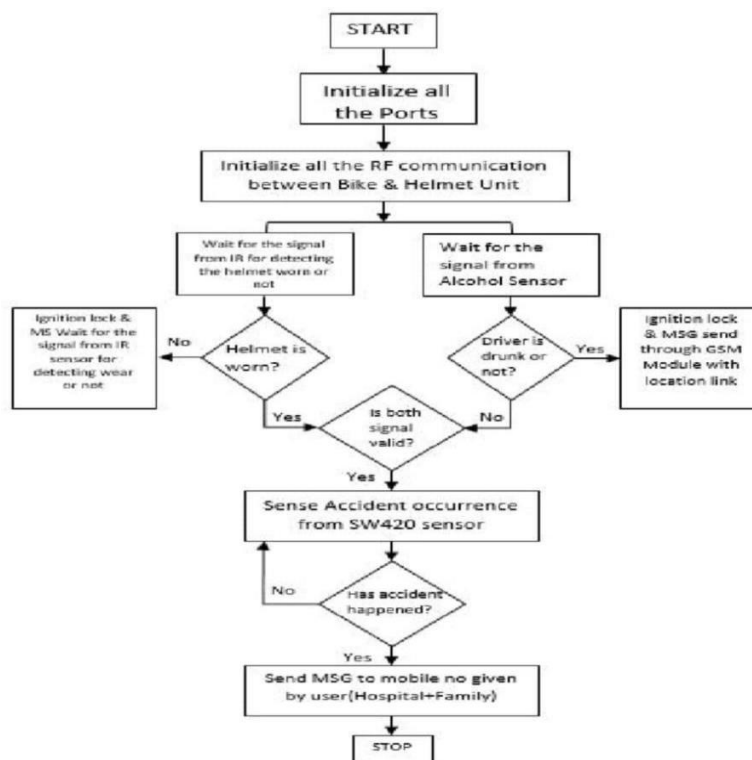


Fig 4.1 System Architecture Of Smart Helmet

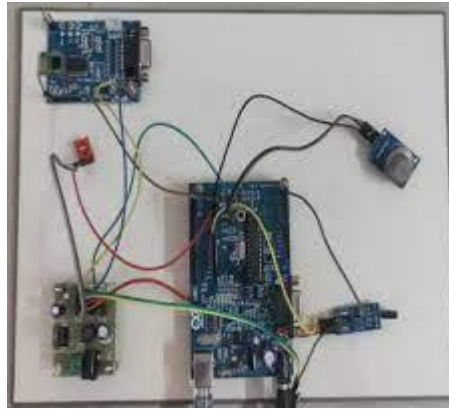


Fig 4.2 (a) Helmet section (Transmitter)



Fig 4.2 (b) Vehicle section (Receiver)

4.1.2 WORKING PRINCIPLE

- Smart helmet system is designed to enhance safety of motorcyclists by incorporating the following key features:
- **Alcohol Detection:** An alcohol sensor (e.g., MQ-3) is utilised for detecting presence of alcohol in the breath of the rider. If alcohol is detected above a certain threshold, the system triggers a warning.
- **Helmet Usage Detection:** A switch is embedded inside the helmet to ensure that the rider is wearing the helmet. The bike ignition system will only be activated if the helmet is worn.
- **Accident Detection:** It could expanse more to include an accelerometer to detect accidents. In the event of an accident, system automatically send message with the rider's location to pre-defined contacts.
- **Location Tracking:** The GPS module provides real-time location data, which is crucial for sending accurate location information if an emergency.

- **Alert System:** The buzzer acts like immediate alert mechanism, sounding off if alcohol is detected or if the helmet is not worn. Additionally, GSM module sends alert to emergency contacts with the location coordinates in case of an accident.

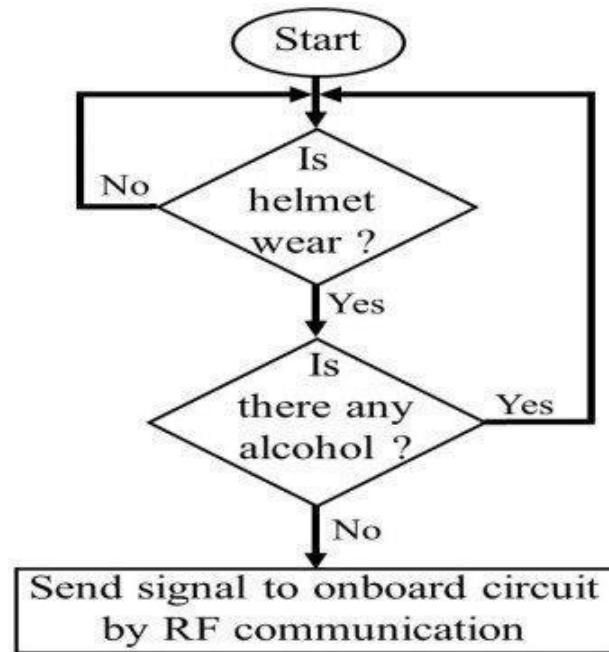


Fig 4.3 Working principle of helmet

4.1.3 THE WORKFLOW DIAGRAM OF PROPOSED SYSTEM

1. Helmet Detection:

- When the rider wears the helmet, the switch is pressed, signaling the Arduino that the helmet is in use.

2. Alcohol Detection:

- Once the helmet is detected, the alcohol sensor continuously monitors the rider's breath.
- If alcohol is detected beyond the set threshold, the buzzer sounds an alarm and the bike's ignition is disabled.

3. System Activation:

- If no alcohol is detected and the helmet is worn, the system allows the bike ignition to start.

4. Accident Detection (Optional):

- The accelerometer detects any sudden impacts. If an accident is detected, the GPS module retrieves the location data.
- The GSM module sends an SMS with the location coordinates to emergency contacts.

5. Location Tracking and Alert:

- The GPS module continuously updates the rider's location.
- In case of emergency, the exact location is sent via SMS to pre-defined contacts.

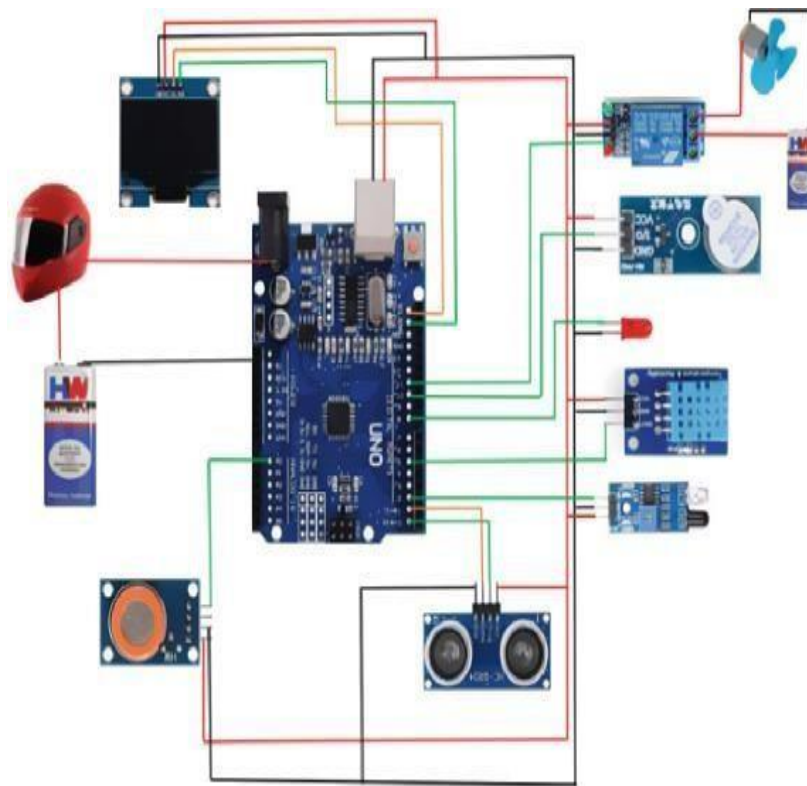


Fig 4.4 Workflow Diagram Of Proposed System

CHAPTER 5

IMPLEMENTATION

5.1 PROJECT IMPLEMENTATION

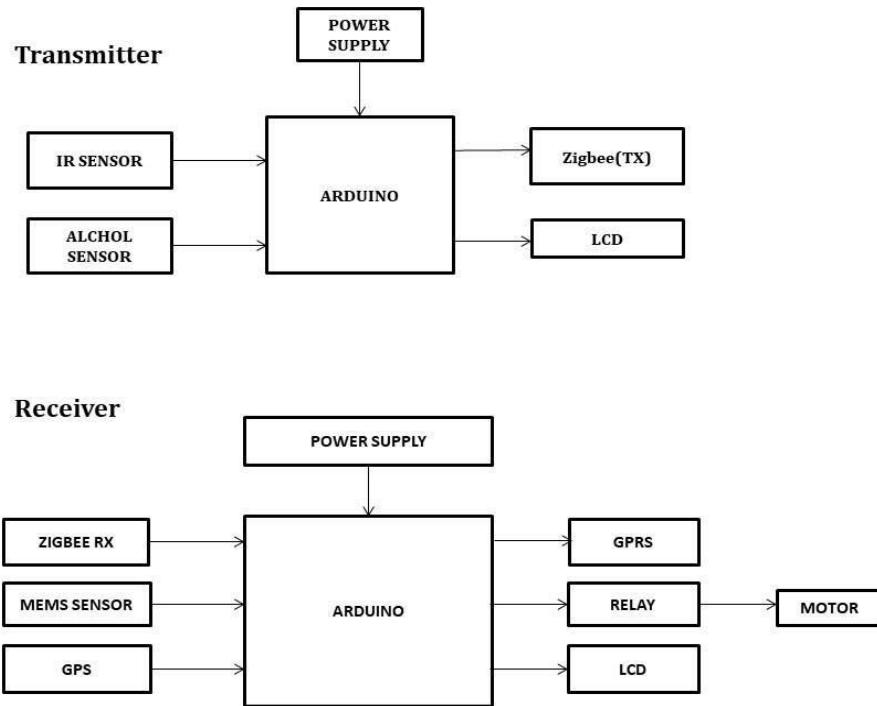


Fig 5.1: implementation diagram of Smart Helmet

Implementing it using a transmitter-based system involves creating a helmet which could wirelessly connect with a receiver placed on the vehicle to control its ignition based on the rider's sobriety. Here's a detailed step-by-step guide to achieve this:

1. Project Overview

Objective: To design our system which could sense alcohol levels in the rider's breath and transmit this data wirelessly to a vehicle's ignition system, preventing the vehicle from starting if alcohol is detected.

Key Components: Helmet with an alcohol sensor, transmitter module, vehicle with a receiver module, microcontroller, and relay for ignition control.

2. System Components

- **Microcontroller:** Arduino, ESP8266, or similar microcontroller to process the sensor data and handle communication
- **Alcohol Sensor:** MQ-3 or MQ-135 sensor to detect alcohol in the rider's breath.
- **Transmitter Module:** RF module, Bluetooth, or other wireless communication module to send data from the helmet to the vehicle.
- **Receiver Module:** Corresponding receiver module on the vehicle side to receive data from the helmet.
- **Relay Module:** Connected to the vehicle's ignition system to control starting based on the received data.
- **Power Supply:** Battery to power the helmet's electronics.
- **Helmet:** A standard motorcycle helmet with space to embed the electronics.
- **Buzzer/LED (Optional):** For alerts and feedback to the rider.

3. Circuit Design for Transmitter (Helmet)

- **Alcohol Sensor:** Connect the MQ-3 sensor to the microcontroller. The analog output of the sensor will be read by one of the microcontroller's analog input pins.
- **Transmitter Module:** Connect the wireless module (e.g., Bluetooth, RF) to the microcontroller. For RF modules, you'll typically connect the data pin to a digital I/O pin on the microcontroller.
- **Power Supply:** Ensure the microcontroller and sensors are powered by a small, portable battery pack. Make sure the voltage is appropriate for all components.
- **Buzzer/LED (Optional):** Connect to another I/O pin to provide feedback.

4. Circuit Design for Receiver (Vehicle)

- **Receiver Module:** The corresponding receiver module will be connected to a microcontroller on the vehicle side.
- **Microcontroller:** This will process the received signal and decide whether to allow the vehicle to start.
- **Relay Module:** Connected to the vehicle's ignition system, the relay will control the ignition based on the microcontroller's output.

5. Programming the Microcontroller

- **Helmet (Transmitter Side):**

Write code to continuously read alcohol levels from the sensor.

If alcohol is detected beyond a set threshold, send a signal via the transmitter module to the receiver. Optionally, trigger a buzzer or LED for rider feedback.

- **Vehicle (Receiver Side):**

Write code to receive the data from the transmitter.

If alcohol is detected, activate the relay to prevent the vehicle from starting.

If no alcohol is detected, allow the vehicle to start normally.

6. Prototyping and Testing

- **Helmet Side:** Test the alcohol sensor for accuracy and the reliability of the wireless transmission.
- **Vehicle Side:** Test the receiver's ability to correctly interpret signals from the helmet and control the relay. Conduct tests under various conditions to ensure that the system reliably detects alcohol and prevents the vehicle from starting if necessary.

7. Integration and Final Assembly

- **Helmet Integration:** Embed all electronic components inside the helmet. Ensure that the design is compact, comfortable, and safe for the rider. **Vehicle Integration:** Secure the receiver and relay module near the vehicle's ignition system, ensuring proper insulation and safety.

8. Deployment

Deploy the system in a real-world scenario and observe its performance.

Collect feedback from users and perform any necessary adjustments to improve the system.

9. Documentation

Document all aspects of the project, including the design, circuit diagrams, code, testing procedures, and results. Prepare a report or presentation to showcase the project's impact and functionality.

10. Potential Enhancements

IoT Integration: Implement an IoT platform to monitor the helmet's data remotely in real-time.

- **Enhanced Security:** Add additional security features, such as password protection or biometric authentication, to prevent tampering.
- **Battery Management:** Implement energy-efficient designs and possibly include solar charging options.

5.2 METHODOLOGY

The smart helmet system for alcohol detection project consists of 5 parts:

- Helmet Authentication for assuring rider is wearing helmet or not.
- Alcohol Detection for assuring rider hasn't consumed alcohol (using MQ3 Gas Sensor).
- Response System in the helmet, for informing bike rider's family regarding accident using GPS & GSM module Helmet.
- Fall Detection-, for informing rider's family about the accident and the vibrations because of stationary hindrances will be detected by using accelerometer.

CHAPTER 6

SYSTEM TESTING

6.1 TEST APPROACH

The test approach for the Arduino-based smart helmet project involves a series of structured steps to validate the functionality, reliability, and performance of the system. The primary focus is on ensuring that each component operates correctly and that the integrated system meets the safety requirements. The testing is divided into unit testing for individual components and system testing for the overall integration.

6.1.1 TESTING PHASES

1. **Unit Testing:** Each component (alcohol sensor, switch, buzzer, GSM module, GPS module) is tested individually to ensure it functions as expected.
2. **Integration Testing:** Components are integrated, and interactions amongst them are put to test for verifying seamless operation.
3. **System Testing:** The entire system is tested in real-world scenarios to ensure overall functionality and performance.
4. **Acceptance Testing:** Final testing is conducted to validate the system against the initial requirements and specifications.

6.2 FEATURES TO BE TESTED

a. Alcohol Detection:

- i. Correct detection of alcohol presence above the threshold.
- ii. Proper activation of the buzzer and disabling of the ignition when alcohol is detected.

b. Helmet Usage Detection:

- i. Accurate detection of helmet usage via the switch.
- ii. Enabling ignition only when the helmet is worn.

c. Accident Detection (if implemented):

- i. Reliable detection of sudden impacts or accidents.
- ii. Correct retrieval of location data from the GPS module.

iii. Prompt sending of SMS alerts with location coordinates via the GSM module.

d. Location Tracking:

- i. Accurate real-time location data from the GPS module.
- ii. Correct formatting and sending of location data in SMS alerts.

e. Alert System:

- i. Audible alerts from the buzzer when alcohol is detected or helmet is not worn.
- ii. Timely sending of SMS alerts to pre-defined contacts in case of emergency.

6.3 TESTING PROCEDURE

1. Unit Testing:

Alcohol Sensor: Connect the alcohol sensor to the Arduino and simulate various breath samples with different alcohol concentrations. Verify that the sensor outputs a signal when the concentration exceeds the threshold. Check the corresponding action (buzzer activation and ignition disable)

Switch: Test the switch by pressing and releasing it to simulate wearing and removing the helmet. Ensure that the ignition is only enabled when the switch is pressed.

Buzzer: Activate the buzzer through the Arduino and check the sound output.

Ensure that the buzzer sounds for alcohol detection and helmet not worn scenarios.

GSM Module: Send test SMS messages via the GSM module. Verify the accuracy and timeliness of the messages.

GPS Module: Retrieve location data via GPS module. Ensure dataset is accurate and correctly formatted for SMS alerts.

2. Integration Testing:

Connect all components to the Arduino and upload the integrated code. Simulate different scenarios (helmet worn, alcohol detection, accident detection) and verify the interaction between components. Ensure that the system responds correctly in each scenario.

3. System Testing:

Scenario 1: Helmet worn, no alcohol detected. Wear the helmet and ensure the switch is pressed. Verify that the ignition is enabled and no alarms are triggered.

Scenario 2: Helmet not worn. Do not wear the helmet and ensure the switch is not pressed. Verify that the ignition is disabled and the buzzer sounds an alert.

Scenario 3: Alcohol detected. Wear the helmet and simulate alcohol detection by breathing into the sensor. Verify that the buzzer sounds an alert and the ignition is disabled.

Scenario 4: Accident detection (if implemented). Simulate an accident by triggering the accelerometer. Verify that the GPS module retrieves the location data and the GSM module sends an SMS alert with the coordinates.

4. Acceptance Testing:

Conduct a final round of testing to ensure all features work as expected. Validate the system against the initial requirements and specifications. Ensure the system is reliable, robust, and meets the safety standards.

CHAPTER 7

RESULT INTERPRETATION AND DISCUSSION

SNAPSHOT 1: Smart Helmet

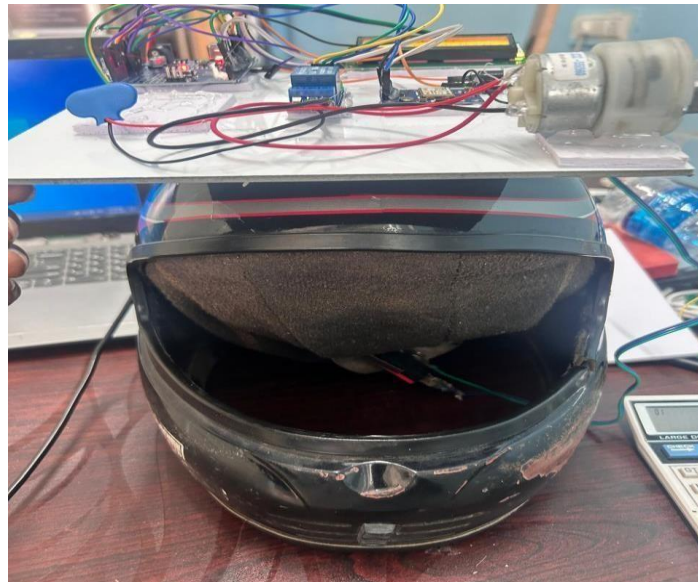


Fig 7.1 System Setup

SNAPSHOT 2: Overall Circuit of System Setup

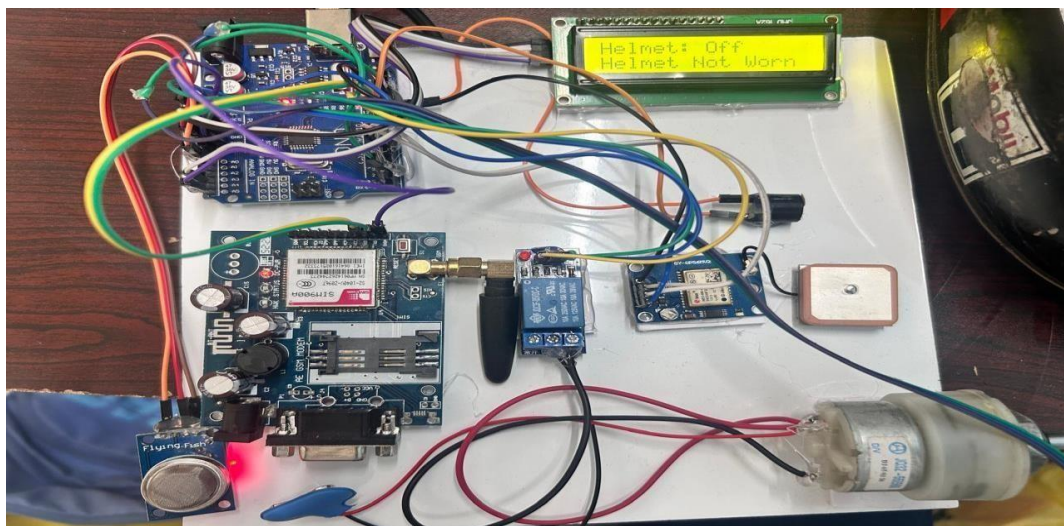


Fig 7.2 Overall Circuit of System Setup

If intoxicated rider attempts to take wheel, bike's alcohol sensors will go into overdrive while disable ignition system, alarm to warn onlookers, and transmit an alert to predetermined cell phone with rider's precise location via GSM while GPS.

SNAPSHOT 3: Helmet Detection

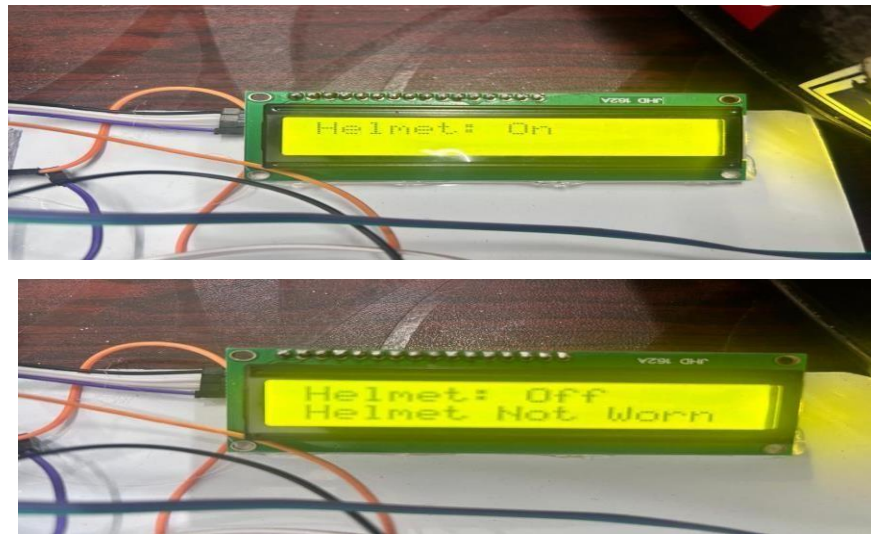


Fig 7.3 Helmet Detection

SNAPSHOT 4: Accident Detection

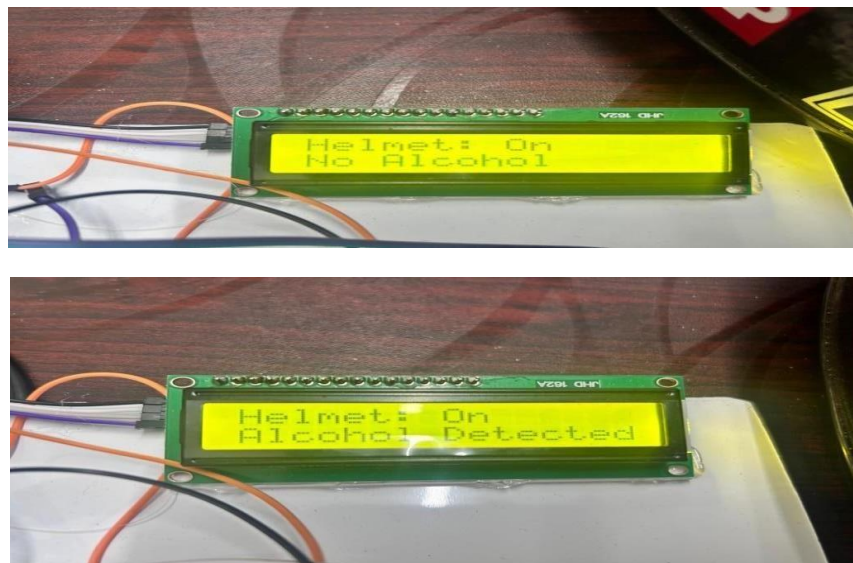


Fig 7.4 Accident Detection

The LCD screen present in the bike will display "Alcohol Detected" so that people are aware of the situation and hence can take the necessary action that may be required. Therefore, by using this system on a bike, any kind of loss of life or damage to property can be avoided. Simulation of the system has been done in Proteus software. All the components have been tested and connected as required there by providing us with the desired result as shown in the above image.

SNAPSHOT 5: Alert Message

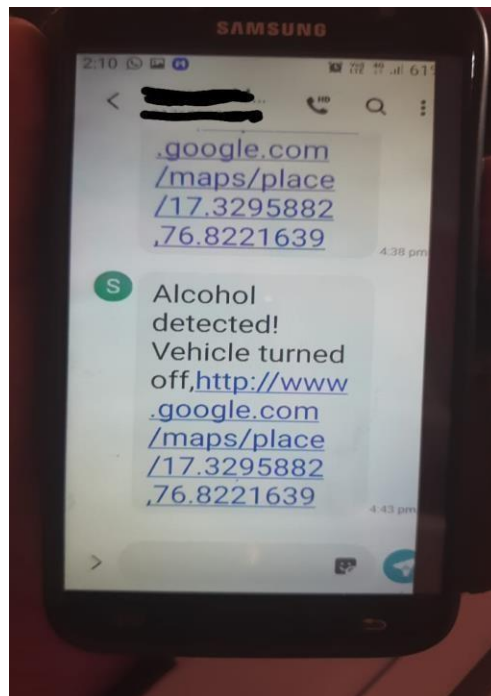


Fig 7.5 Alert message on the phone after the accident

CONCLUSION

Smart helmet system for alcohol detection can decrease number of street mishaps that happens consistently. It guarantees the security of the biker as well as sends the casualty's area to relatives and close by police headquarters. Additionally, demise rate can definitely be diminished by carrying out this circuit as compulsory while driving and make everybody's life more straight-forward and smoother. Our project undertaking addresses an illustration of efficient for dealing individuals from mishaps. In this manner simply by wearing the protective helmet we can begin the bicycle. By smart helmet system we can keep away from numerous mishaps in this world, a large portion of the mishaps are happening a direct result of not wearing helmet, our administration is acquainting part of rules with wear helmet yet our kind isn't answering for that, so this is a superior method for observing the guidelines to stay away from mishap and save individuals from the weighty head injury.

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